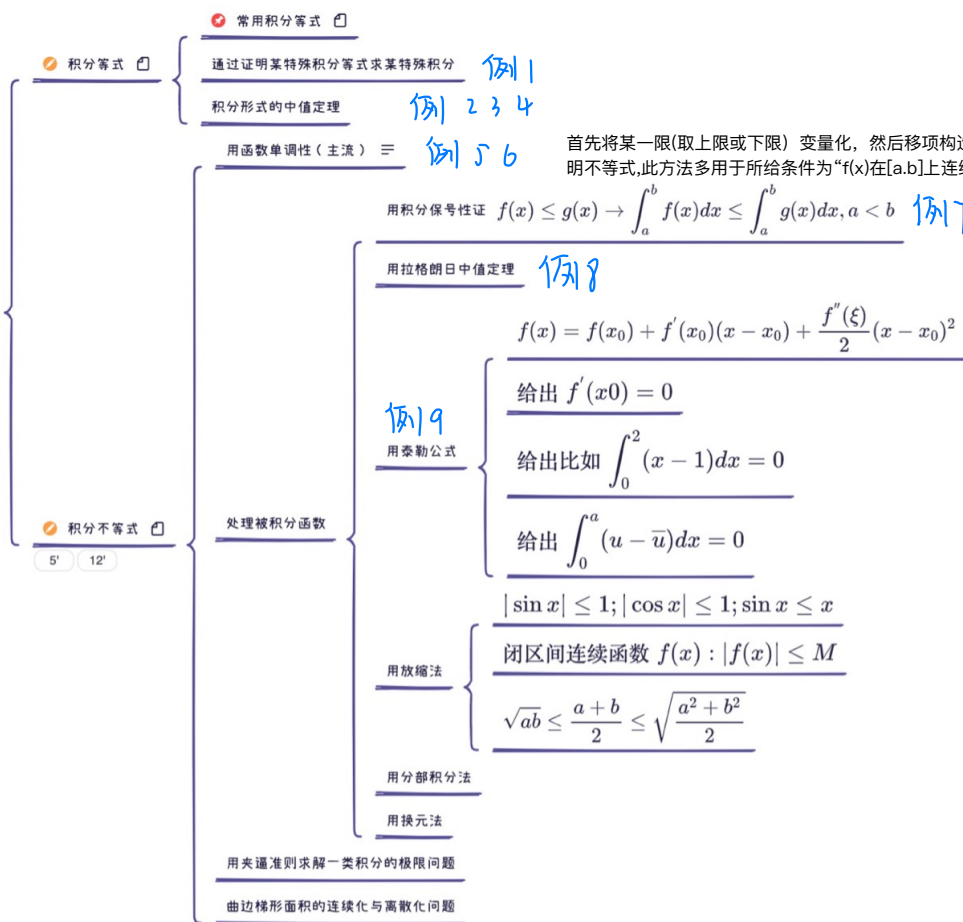


高等数学

积分等式与积分不等式 (证明题)



考试会有提示,「设置第一问」; 各种方法综合出现

例 11.1

例 11.1 (1) 设 $f(x)$ 是 $[-a, a] (a > 0)$ 上的连续偶函数, 证明 $\int_{-a}^a \frac{f(x)}{1-e^x} dx = \int_0^a f(x) dx$;

(2) 计算 $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^3 x}{1-e^x} dx$.

$$\begin{aligned} (1) & \int_{-a}^0 \frac{f(x)}{1-e^x} dx + \int_0^a \frac{f(x)}{1-e^x} dx \\ &= \int_0^a \frac{f(-x)}{1-e^{-x}} + \frac{f(x)}{1-e^x} dx \\ &= \int_0^a f(x) \left[\frac{1}{1-e^{-x}} + \frac{1}{1-e^x} \right] dx = \frac{1-e^x + 1-e^{-x}}{(1-e^{-x})(1-e^x)} = 1 \\ &= \int_0^a f(x) dx \end{aligned}$$

$$(2) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^3 x}{1-e^x} dx = \int_0^{\frac{\pi}{2}} \sin x dx = \frac{1}{6} \frac{1}{4} \frac{1}{2} \frac{\pi}{2}$$

例 11.2

例 11.2 设 $f(x)$ 在区间 $[-1, 1]$ 上连续, 且 $\int_{-1}^1 f(x) dx = \int_{-1}^1 f(x) \tan x dx = 0$, 证明在区间 $(-1, 1)$ 内至少存在互异的两点 ξ_1, ξ_2 , 使得 $f(\xi_1) = f(\xi_2) = 0$.

$$F(x) = \int_{-1}^x f(t) dt \Rightarrow F(-1) = 0 \quad F(1) = 0$$

反证法 设 $F(x) > 0 \quad x \in (-1, 1)$

$$\text{则 } \int_{-1}^1 f(x) \tan x dx = \int_{-1}^1 \tan x dF(x) = \tan x F(x) \Big|_{-1}^1 - \int_{-1}^1 F(x) d \tan x = - \int_{-1}^1 F(x) \sec^2 x dx < 0$$

矛盾 $\Rightarrow$ 存在 $F(x_0) = 0 \quad x_0 \in (-1, 1)$

$$\frac{F(x_0) - F(-1)}{x_0 - (-1)} = f(\xi_1) = 0 \quad \frac{F(1) - F(x_0)}{1 - x_0} = f(\xi_2) = 0$$

例 11.3

例 11.3 设 $f(x), g(x)$ 在 $[a, b]$ 上连续且 $g(x)$ 不变号, 证明: 至少存在一点 $\xi \in [a, b]$,

使得

$$\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx.$$

当 $g(x) \equiv 0$ 时 成立

当 $g(x) > 0$ 时 $\int_a^b g(x)dx > 0$ $f(x)$ 在 $[a, b]$ 有界
 $\Rightarrow m \leq f(x) \leq M$

$$\Rightarrow m g(x) \leq f(x) g(x) \leq M g(x)$$

$$m \int_a^b g(x)dx \leq \int_a^b f(x)g(x)dx \leq M \int_a^b g(x)dx$$

$$m \leq \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx} \leq M$$

$$\text{介值定理: } \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx} = f(\xi)$$

例 11.4

例 11.4 设 $f(x), g(x)$ 在 $[a, b]$ 上连续, 且 $g(x)$ 在 $[a, b]$ 上不变号, 证明: $\exists \xi \in (a, b)$, 使

$$\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx.$$

$g(x) \equiv 0$ 时 成立

不妨设 $g(x) > 0$ 时

$$F(x) = \int_a^x f(t)g(t)dt$$

$$G(x) = \int_a^x g(t)dt$$

$$\frac{F(x) - F(a)}{G(x) - G(a)} = \frac{F'(x)}{G'(x)} = \frac{\int_a^x f(t)g(t)dt}{\int_a^x g(t)dt} = f(\xi)$$

例 11.5

例 11.5 设 $f(x)$ 在 $[0, 1]$ 上可导, $0 \leq f'(x) \leq 1$ 且 $f(0) = 0$, 证明:

$$\left[\int_0^1 f(x)dx \right]^2 \geq \int_0^1 f^3(x)dx.$$

$$F(x) = \left(\int_0^x f(t)dt \right)^2 - \int_0^x f^3(t)dt$$

$$F'(x) = 2 \left(\int_0^x f(t)dt \right) \cdot f(x) - f^3(x) = f(x) \left[2 \int_0^x f(t)dt - f^2(x) \right] = G(x)$$

$$F(0) = 0$$

$$G'(x) = 2f(x) - 2f(x) \cdot f(x)$$

$$\left. \begin{array}{l} f(0) = 0 \\ f(x) \in (0, 1) \end{array} \right\} \Rightarrow f(x) > 0 \Rightarrow G'(x) = 2f(x)(1 - f(x)) > 0 \Rightarrow G(x) \uparrow \Rightarrow G(x) > 0$$

$$\Rightarrow F'(x) > 0 \quad F(x) \uparrow \quad F(0) = 0 \quad *$$

例 11.6

例 11.6 设 $f(x)$ 在 $[a, b]$ 上单调增加且连续, 证明: $\int_a^b x f(x)dx \geq \frac{a+b}{2} \int_a^b f(x)dx$.

$$F(x) = \int_a^x t f(t)dt - \frac{a+x}{2} \int_a^x f(t)dt \quad \text{证 } F(b) \geq 0$$

$$F'(x) = x f(x) - \frac{1}{2} \int_a^x f(t)dt - \frac{a+x}{2} f(x)$$

$$= 2 \left[(x-a)f(x) - \int_a^x f(t)dt \right]$$

$$= 2 \int_a^x (f(x) - f(t))dt \geq 0$$

$$F(x) \uparrow \quad F(a) = 0$$

例 11.7

例 11.7 设 $f(x)$ 为正值连续函数且 $f(x) < a$, a 为正常数, 则 $\forall b \in (0, 1)$, 有 ().

- (A) $a \int_0^1 \sqrt{f(bx)} dx < \sqrt{b}$ (B) $a \int_0^1 \sqrt{f(bx)} dx < b$
(C) $b \int_0^1 \sqrt{f(bx)} dx < \sqrt{a}$ (D) $b \int_0^1 \sqrt{f(bx)} dx < a$

$$\int_0^1 \sqrt{f(bx)} dx \xrightarrow{bx=t} \int_0^b \sqrt{f(t)} \frac{dt}{b} < \int_0^1 \frac{\sqrt{a}}{b} dt = \frac{\sqrt{a}}{b} \Rightarrow C$$

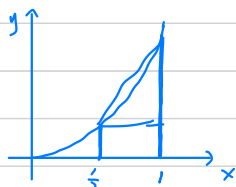
例 11.8

例 11.8 设 $f(x)$ 在 $[0, 1]$ 上二阶可导, $f(0) = 0$, $f'(x) > 0$, $f''(x) > 0$, 则对于

$$M = \int_0^{\frac{1}{2}} f(x) dx, N = \int_{\frac{1}{2}}^1 [f(x) - f(\frac{1}{2})] dx, P = \frac{1}{4} [f(1) - f(\frac{1}{2})].$$

其大小顺序排列正确的是 ().

- (A) $N < M < P$ (B) $P < M < N$
(C) $M < P < N$ (D) $M < N < P$



$$\begin{aligned} \textcircled{1} N - M &= \int_{\frac{1}{2}}^1 f(x) dx - f(\frac{1}{2}) \int_{\frac{1}{2}}^1 dx - \int_0^{\frac{1}{2}} f(x) dx \\ &= \int_0^{\frac{1}{2}} [f(x+\frac{1}{2}) - f(\frac{1}{2})] dx - \int_0^{\frac{1}{2}} f(x) dx \\ &= \int_0^{\frac{1}{2}} (f(x+\frac{1}{2}) - f(\frac{1}{2}) - f(x)) dx \\ &= \int_0^{\frac{1}{2}} [f(x+\frac{1}{2}) - f(\frac{1}{2}) - (f(x) - f(0))] dx \\ &= \int_0^{\frac{1}{2}} [f'(y_1) - f'(\frac{1}{2})] x dx > 0 \quad \because f''(x) > 0 \Rightarrow f'(x) \uparrow \end{aligned}$$

$$\textcircled{2} N = \int_{\frac{1}{2}}^1 [f(x) - f(\frac{1}{2})] dx$$

$$P = \frac{1}{4} [f(1) - f(\frac{1}{2})]$$

$$P > N > M \quad (D)$$

若 $f'(x) > 0$, $f''(x) > 0$,
 证: $\frac{1}{2} [f(a) + f(b)](b-a) > \int_a^b f(x) dx$
 令 $F(x) = \frac{1}{2} [f(a) + f(x)](x-a) - \int_a^x f(t) dt > 0$
 $F'(x) = \frac{1}{2} [f(a) + f(x)] + \frac{1}{2} f'(x)(x-a) - f(x)$
 $= \frac{1}{2} (f(a) - f(x)) + \frac{1}{2} f'(x)(x-a)$
 $= \frac{1}{2} (x-a) (f'(x) - f'(y)) > 0 \quad f'(x) \uparrow$
 $F(a) = 0 \quad F(b) > 0$

例 11.9

例 11.9 设 $f(x)$ 二阶可导, 且 $f''(x) \geq 0$, $u(t)$ 为任一连续函数, $a > 0$. 证明:

$$\frac{1}{a} \int_0^a f(u(t)) dt \geq f\left[\frac{1}{a} \int_0^a u(t) dt\right].$$

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\xi)}{2} (x-x_0)^2 \\ &\geq f(x_0) + f'(x_0)(x-x_0) \end{aligned}$$

$$\frac{1}{a} \int_0^a f(u(t)) dt \geq \frac{1}{a} \int_0^a [f(x_0) + f'(x_0)(u(t)-x_0)] dx = \frac{1}{a} \int_0^a f(x_0) dx$$

$$\text{其中 } \int_0^a [u(t) - \frac{1}{a} \int_0^a u(t) dt] dt = \int_0^a u(t) dt - \int_0^a u(t) dt = 0$$

例 11.10

例 11.10 设函数 $f(x) = \int_x^{x+1} \sin e^t dt$, 证明:

$$(1) f(x) = \frac{\cos e^x}{e^x} - \frac{\cos e^{x+1}}{e^{x+1}} - \int_{e^x}^{e^{x+1}} \frac{1}{u^2} \cos u du;$$

$$(2) e^x |f(x)| \leq 2.$$

$$(1) f(x) = \int_x^{x+1} \sin e^t dt \xrightarrow{t=\ln u} \int_{e^x}^{e^{x+1}} \sin u \frac{du}{u} = \int_{e^x}^{e^{x+1}} \frac{-1}{u} d \cos u = \left. \frac{-\cos u}{u} \right|_{e^x}^{e^{x+1}} - \int_{e^x}^{e^{x+1}} \frac{\cos u}{u^2} du$$

$$f(x) = \frac{\cos e^x}{e^x} - \frac{\cos e^{x+1}}{e^{x+1}} - \int_{e^x}^{e^{x+1}} \frac{1}{u^2} \cos u du$$

$$(2) e^x |f(x)| \leq \left| \frac{\cos e^x}{e^x} \right| + \left| \frac{\cos e^{x+1}}{e^{x+1}} \right| + e^x \cdot \int_{e^x}^{e^{x+1}} \frac{1}{u^2} du$$

$$\leq \frac{1}{e^x} + \frac{1}{e^{x+1}} + e^x \int_{e^x}^{e^{x+1}} \frac{1}{u^2} du$$

$$= \frac{1}{e^x} - \frac{1}{e^{x+1}} + 1 = 2$$

例 11.11

例 11.11 设 $f(x) = \int_x^{x+1} \sin t^2 dx, x > 0$, 证明:

$$(1) f(x) = \frac{1}{2} \left[\frac{\cos x^2}{x} - \frac{\cos (x+1)^2}{x+1} \right] - \frac{1}{2} \int_x^{x+1} \frac{\cos t^2}{t^2} dt;$$

$$(2) |f(x)| \leq \frac{1}{x}.$$

$$(1) (\cos t^2)' = -\sin t^2 \cdot 2t \quad dt$$

$$f(x) = \int_x^{x+1} -\frac{1}{2t} d \cos t^2 = -\frac{\cos t^2}{2t} \Big|_x^{x+1} - \frac{1}{2} \int_x^{x+1} \frac{\cos t^2}{t^2} dt$$

$$f(x) = \frac{1}{2} \left[\frac{\cos x^2}{x} - \frac{\cos (x+1)^2}{x+1} \right] - \frac{1}{2} \int_x^{x+1} \frac{\cos t^2}{t^2} dt$$

$$(2) |f(x)| \leq \frac{1}{2} \left[\left| \frac{\cos x^2}{x} \right| + \left| \frac{\cos (x+1)^2}{x+1} \right| + \int_x^{x+1} \frac{1}{t^2} dt \right]$$

$$\leq \frac{1}{2} \left[\frac{1}{x} + \frac{1}{x+1} + \int_x^{x+1} \frac{1}{t^2} dt \right]$$

$$= \frac{1}{x}$$



例 11.12

$$\frac{1}{g'(t)} \geq m$$

例 11.12 设 $|f(x)| \leq \pi, f'(x) \geq m > 0 (a \leq x \leq b)$, 证明 $\left| \int_a^b \sin f(x) dx \right| \leq \frac{2}{m}$.



$$f(x)=t \Rightarrow x=f^{-1}(t)=g(t) \Rightarrow 0 < g'(t) \leq \frac{1}{m}$$

$$\left| \int_{g(a)}^{g(b)} \sin t \cdot g'(t) dt \right| \leq \left| \frac{1}{m} \int_a^b \sin t dt \right| \leq \frac{1}{m} \left| \int_0^\pi \sin t dt \right| = \frac{2}{m}$$



例 11.13

例 11.13 设 $f(n) = \int_0^{\frac{\pi}{4}} \tan^n x dx, n = 3, 4, \dots$.

$$(1) \text{ 证明 } f(n) + f(n+2) = \frac{1}{1+n};$$

$$(2) \text{ 计算 } \lim_{n \rightarrow \infty} n f(n).$$

$$(1) f(n) + f(n+2) = \int_0^{\frac{\pi}{4}} \tan^n x (1 + \tan^2 x) dx = \int_0^{\frac{\pi}{4}} \tan^n x d \tan x = \left. \frac{\tan^{n+1} x}{n+1} \right|_0^{\frac{\pi}{4}} = \frac{1}{n+1}$$

$$(2) \begin{cases} f(n-1) + f(n+1) = \frac{1}{n} \\ f(n-2) + f(n) = \frac{1}{n-1} \end{cases} \quad \lim_{n \rightarrow \infty} n f(n) = \frac{1}{2}$$

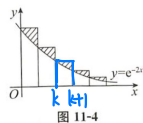
$$f(n) + f(n+2) < 2f(n) < f(n-2) + f(n)$$

$$\frac{1}{2} \frac{n}{1+n} = n \frac{f(n) + f(n+1)}{2} < n f(n) < n \frac{f(n-1) + f(n)}{2} = \frac{1}{2} \frac{n}{n-1}$$

$\downarrow$ $\downarrow$
 $\downarrow$ $\downarrow$
 $\downarrow$ $\downarrow$

例 11.14

例 11.14 当 $x \geq 0$ 时, 在曲线 $y = e^{-2x}$ 上面作一个台阶曲线, 台阶的宽度皆为 1 (见图 11-4). 求图中无穷多个阴影部分的面积之和 S .



$$S_k = e^{-2k} - \int_k^{k+1} e^{-2x} dx = e^{-2k} - \frac{e^{-2x}}{-2} \Big|_k^{k+1} = \frac{1}{2} (e^{-2(k+1)} + e^{-2k})$$

$$S = \frac{1}{2} \sum_{k=0}^{\infty} e^{-2(k+1)} + e^{-2k} = \frac{1}{2} \left(\frac{e^{-2}}{1-e^{-2}} + \frac{1}{1-e^{-2}} \right) = \frac{e^2+1}{2(e^2-1)}$$



习 11.1

11.1 设 $f(x)$ 是连续的偶函数, 且是以 T 为周期的周期函数.

(1) 证明 $\int_0^{nT} x f(x) dx = \frac{n^2 T}{2} \int_0^T f(x) dx$ ($n = 1, 2, 3, \dots$);

(2) 利用 (1) 的结论计算 $I = \int_0^{n\pi} x |\sin x| dx$.

$$\begin{aligned}
 (1) \quad \int_0^{nT} x f(x) dx &= \int_0^T x f(x) dx + \int_T^{2T} x f(x) dx + \dots + \int_{(k-1)T}^{kT} x f(x) dx + \dots \\
 &= \int_0^T x f(x) dx + \int_0^T (x-T) f(x) dx + \dots + \int_0^T (x-(k-1)T) f(x) dx + \dots \\
 &= n \int_0^T x f(x) dx - \frac{(0+n-1)n}{2} \int_0^T T f(x) dx \\
 &= n \int_0^T x f(x) dx - \frac{n(n-1)T}{2} \int_0^T f(x) dx \\
 &\stackrel{x=T-t}{=} -n \int_0^T t f(t) dt - \frac{n(n-1)T}{2} \int_0^T T f(t) dt
 \end{aligned}$$

$$(1) \quad \int_0^{nT} x f(x) dx = \int_0^{nT} (nT-x) f(nT-x) dx = \int_0^{nT} (nT-x) f(x) dx$$

$$\Rightarrow \int_0^{nT} x f(x) dx = \frac{nT}{2} \int_0^{nT} f(x) dx = \frac{n^2 T}{2} \int_0^T f(x) dx$$



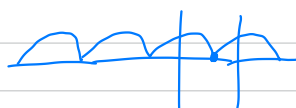
$$(2) \quad I = \int_0^{n\pi} x |\sin x| dx = \frac{n\pi}{2} \int_0^{n\pi} |\sin x| dx = n\pi$$

习 11.2

11.2 设 $f(x)$ 是周期为 2 的连续函数. 证明:

(1) 对任意的实数 t , 有 $\int_t^{t+2} f(x) dx = \int_0^2 f(x) dx$;

(2) $G(x) = \int_0^x [2f(t) - \int_t^{t+2} f(s) ds] dt$ 是周期为 2 的周期函数.



$$\begin{aligned}
 (1) \quad \int_t^{t+2} f(x) dx &= \int_t^{2k} f(x) dx + \int_{2k}^{t+2} f(x) dx = \int_{2k}^{t+2} f(x) dx + \int_{t+2}^{2k+2} f(x) dx \\
 &= \int_{2k}^{2k+2} f(x) dx = \int_0^2 f(x) dx
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad G(x+2) - G(x) &= \int_x^{x+2} [2f(t) - \int_t^{t+2} f(s) ds] dt \\
 &\stackrel{u=t-x}{=} \int_0^2 [2f(u+x) - \int_{u-x}^{u-x+2} f(s) ds] du
 \end{aligned}$$

$$= \int_0^2 [2f(u-x) - \int_0^2 f(s) ds] du$$

$$G(2) - G(0) = \int_0^2 [2f(t) - \int_t^{t+2} f(s) ds] dt = \int_0^2 [2f(t) - \int_0^2 f(s) ds] dt$$

$$\Rightarrow G(x+2) - G(x) = G(2) - G(0)$$

→ $G'(x)$ 以 2 为周期

$$G(x) = \int_0^x [2f(t) - \int_0^2 f(s)ds] dt = 2 \int_0^x f(t) dt - \int_0^1 (\int_0^2 f(s)ds) dt$$

$$= 2 \int_0^x f(t) dt - 2 \int_0^1 f(s)ds = 0$$

⇒ $G'(x)$ 在一个周期上积分为 0

⇒ $G(x)$ 以 2 为周期

例 11.3

11.3 设 $f(x), g(x)$ 在 $[0, 1]$ 上的导数连续, 且 $f(0) = 0, f'(x) \geq 0, g'(x) \geq 0$. 证明对任意的 $a \in [0, 1]$, 有

$$\int_0^a g(x) f'(x) dx + \int_0^1 f(x) g'(x) dx \geq f(a) g(1).$$

$$F(x) = \int_0^x g(t) f'(t) dt + \int_0^1 f(t) g'(t) dt - f(x) g(x)$$

$$F(0) = \int_0^1 f(t) g'(t) dt > 0 \quad F(1) = \int_0^1 g(t) f'(t) dt + \int_0^1 f(t) g'(t) dt - f(1) g(1) = 0$$

$$F'(x) = g(x) f'(x) - f(x) g'(x) = f'(x) (g(x) - g(1))$$

$$\because g'(x) \geq 0 \Rightarrow g(x) \leq g(1) \quad (x \in [0, 1])$$

$$\Rightarrow F'(x) \leq 0 \quad F(x) \downarrow \quad F(1) = 0 \Rightarrow F(x) \geq 0.$$

例 11.4

11.4 设 $f(x)$ 在 $[0, 2]$ 上连续, 在 $(0, 2)$ 内可导, 且 $f(0) = f(2) = 1, |f'(x)| \leq 1$. 证明: $\int_0^2 f(x) dx < 3$.

$$\text{明: } \int_0^2 f(x) dx < 3.$$

$$f(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$f(x) = f(0) + f'(\xi_1)x = 1 + f'(\xi_1)x \quad \xi_1 \in (0, x)$$

$$f(x) = f(2) + f'(\xi_2)(x-2) = 1 + f'(\xi_2)(x-2) \quad \xi_2 \in (x, 2)$$

$$|f(x)| \leq 1 \quad -1 \leq f'(x) \leq 1 \quad f(0) = f(2) = 1$$

$$\Rightarrow \begin{cases} xf'(\xi_1) \leq x & f(x) \leq 1+x \\ (x-2)f'(\xi_2) \leq 2-x & f(x) \leq 3-x \end{cases}$$

$$g(x) = \begin{cases} 1+x, & 0 \leq x \leq 1 \\ 3-x, & 1 \leq x \leq 2 \end{cases} \quad f(x) \leq g(x)$$

$$\int_0^2 f(x) dx \leq \int_0^2 g(x) dx = 3$$

例 11.5

11.5 设 $f(x), g(x)$ 在 $[a, b]$ 上连续, 且满足

$$\int_a^x f(t) dt \geq \int_a^x g(t) dt, x \in [a, b], \int_a^b f(t) dt = \int_a^b g(t) dt,$$

$$\text{证明 } \int_a^b x f(x) dx \leq \int_a^b x g(x) dx.$$

$$F(x) = \int_a^x f(t) - g(t) dt \geq 0$$

$$F'(x) = f(x) - g(x)$$

$$F(a) = 0 \quad F(b) = 0$$

$$G(x) = \int_a^x t[f(t) - g(t)] dt = \int_a^x t dF(t) = tF(t) \Big|_a^x - \int_a^x F(t) dt = - \int_a^x F(t) dt$$

$$G'(x) = -F(x) \leq 0$$

$$G(x) \downarrow \quad G(a) = 0 \Rightarrow G(x) \leq 0$$

例 11.6

11.6 证明: 当 $0 \leq a \leq 1$ 时, $\int_0^a (1-x^2)^{\frac{3}{2}} dx \geq \frac{5a\pi}{32}$.

$$\frac{3.14}{1.5}$$

$$F(x) = \int_0^x (1-t^2)^{\frac{3}{2}} dt - \frac{5x\pi}{32} \quad x \in [0, 1]$$

$$F'(x) = (1-x^2)^{\frac{3}{2}} - \frac{5\pi}{32} \stackrel{?}{\geq} 0 \Rightarrow 1-x^2 \geq \frac{(5\pi)^{\frac{2}{3}}}{4} \quad x_0 = \sqrt{1 - \frac{(5\pi)^{\frac{2}{3}}}{4}}$$

x	0	$(0, x_0)$	x_0	$(x_0, 1)$	1
$F'(x)$	+	+	0	-	-
$F(x)$	0	$\uparrow$	$F(x_0)$	$\downarrow$	0

$$F(x) = \int_0^1 (1-t)^{\frac{1}{2}} dt - \frac{5\pi}{32} \Rightarrow F(x) > 0$$

$$\begin{aligned} t = \sin x &= \int_0^{\frac{\pi}{2}} \cos^6 x dx - \frac{5\pi}{32} \\ &= \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{1}{2} - \frac{5\pi}{32} = 0 \end{aligned}$$

习 11.7

11.7 设 $f(x)$ 在 $[0, 2\pi]$ 上具有一阶连续导数, 且 $f'(x) \geq 0$, 证明: 对任意正整数 n 有

$$\left| \int_0^{2\pi} f(x) \sin nx dx \right| \leq \frac{2}{n} [f(2\pi) - f(0)].$$

证: $\left| \int_0^{2\pi} f(x) \sin nx dx \right| - \frac{2}{n} [f(2\pi) - f(0)]$

$$\begin{aligned} F(x) &= \left| \int_0^x f(t) \sin nt dt \right| - \frac{2}{n} [f(x) - f(0)] + \left| \frac{f(0)}{n} \right| \\ &= \left| \int_0^x f(t) d \frac{\cos nt}{n} \right| - \frac{1}{n} [f(x) - f(0)] \\ &\leq \left| \frac{\cos nx}{n} f(x) \right| + \left| \int_0^x \frac{\cos nt}{n} f'(t) dt \right| - \frac{1}{n} [f(x) - f(0)] \\ &\leq \frac{2f(x)}{n} + \frac{2f(0)}{n} - \frac{2}{n} f(x) - \frac{2}{n} f(0) = 0 \end{aligned}$$

习 11.8

11.8 设 n 为大于 1 的整数, 证明不等式 $(n-1)! < e\left(\frac{n}{e}\right)^n < n!$.

即证 $\sum_{k=1}^{n-1} \ln k < 1 + n \ln n - n < \sum_{k=1}^n \ln k$

$y = \ln x$ 在 $[1, n]$ 上 $\uparrow$

$x \in [k, k+1]$ 时

$\ln k < \ln x < \ln(k+1)$

$$\sum_{k=1}^{n-1} \ln k = \sum_{k=1}^{n-1} \int_k^{k+1} \ln k dx < \sum_{k=1}^{n-1} \int_k^{k+1} \ln x dx$$

右端点 $\uparrow$

$$= \sum_{k=1}^{n-1} \int_k^{k+1} \ln x dx < \sum_{k=1}^{n-1} \int_k^{k+1} \ln(k+1) dx = \sum_{k=1}^{n-1} \ln(k+1)$$

右端点 $\uparrow$

$$= \int_1^n \ln x dx = x \ln x \Big|_1^n - \int_1^n 1 dx = n \ln n - (n-1)$$

习 11.9

11.9 设 $f(x)$ 是区间 $[0, +\infty)$ 内单调减少且非负连续函数,

$$a_n = \sum_{k=1}^n f(k) - \int_1^n f(x) dx \quad (n=1, 2, \dots),$$

证明数列 $\{a_n\}$ 的极限存在.

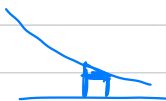
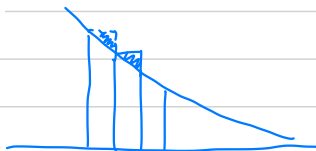
$$a_n = \sum_{k=1}^n f(k) - \int_1^n f(x) dx$$

$$a_n = \sum_{k=1}^{n-1} \int_k^{k+1} f(k) - f(x) dx + f(n) \geq 0$$

$$a_{n+1} - a_n = f(n+1) - f(n) + \int_n^{n+1} f(n) - f(x) dx = f(n+1) - \int_n^{n+1} f(x) dx < 0$$

$\Rightarrow \{a_n\} \downarrow$

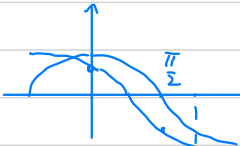
单调有界 $\Rightarrow \{a_n\}$ 极限存在



1000.11.1.1

1. 设常数 $\alpha > 0$, 积分 $I_1 = \int_0^{\frac{\pi}{2}} \frac{\cos x}{1+x^\alpha} dx$, $I_2 = \int_0^{\frac{\pi}{2}} \frac{\sin x}{1+x^\alpha} dx$, 则 ().

(A) $I_1 > I_2$ (B) $I_1 < I_2$ (C) $I_1 = I_2$ (D) I_1 与 I_2 的大小与 α 有关

$$\int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{1+x^\alpha} dx = \int_0^{\frac{\pi}{2}} \frac{\sqrt{2} \cos(x - \frac{\pi}{4})}{1+x^\alpha} dx > 0 \quad (A)$$


1000.11.1.2

2. 设 $f(x)$ 是 $[0,1]$ 上连续且单调减少的正值函数, 则对于任意的 $a, b (0 < a < b < 1)$, 下列结论不正确的是 (A).

(A) $a \int_0^b f(x) dx > b \int_0^a f(x) dx$ (B) $b \int_0^a f(x) dx > a \int_0^b f(x) dx$ (C) $a \int_0^b \sqrt{f(x)} dx < b \int_0^a \sqrt{f(x)} dx$ (D) $b \int_0^a \sqrt{f(x)} dx < a \int_0^b \sqrt{f(x)} dx$

$$f(x) > 0 \quad f'(x) < 0$$

$$A: \frac{1}{b} \int_0^b f(x) dx > \frac{1}{a} \int_0^a f(x) dx$$

$$F(x) = \frac{1}{x} \int_0^x f(t) dt$$

$$F'(x) = -\frac{1}{x^2} \int_0^x f(t) dt + \frac{f(x)}{x} = \frac{x f(x) - \int_0^x f(t) dt}{x^2} < 0$$

$$F(b) < F(a)$$

1000.11.1.3

3. $\lim_{n \rightarrow \infty} \int_{-1}^2 (\arctan nx)^3 dx = \underline{\quad\quad\quad}$.

$$\lim_{n \rightarrow \infty} \int_{-1}^2 (\arctan nx)^3 dx$$

$$= \int_{-\infty}^{+\infty} (\arctan x)^3 dx + (-\frac{\pi}{2})^3 + 2(\frac{\pi}{2})^3$$

$$= \frac{\pi^3}{8}$$

1000.11.1.4

4. 设 $f(x)$ 在 $[a, b]$ 上连续, 且 $f(x) > 0$, 证明: 至少存在一点 $\xi \in (a, b)$, 使得

$$\int_a^b f(x) dx = \int_a^\xi f(x) dx + \int_{a+\xi}^b f(x) dx.$$

$$\int_a^b f(x) dx = \int_a^\xi f(x) dx + \int_\xi^b f(x) dx$$

$$\text{设 } \varepsilon = a+b-\xi \Rightarrow \xi = \frac{a+b}{2}$$

$$\text{则 } \varepsilon = \frac{a+b}{2} \text{ 即可满足}$$

1000.11.1.5

5. 设 $f(x)$ 在 $[a, b]$ 上连续且单调增加, 证明: 至少存在一点 $\xi \in [a, b]$, 使得

$$\int_a^b f(x) dx = f(a)(\xi-a) + f(b)(b-\xi).$$

$$F(x) = f(a)(x-a) + f(b)(b-x) - \int_a^b f(x) dx$$

$$F'(x) = f(a) - f(b) < 0$$

$$F(a) = f(b)(b-a) - \int_a^b f(x) dx = \int_a^b f(b) - f(x) dx > 0$$

$$F(b) = \int_a^b f(a) - f(x) dx < 0$$

$$\Rightarrow F(\xi) = 0$$

1000.11.1.6

6. 设函数 $f(x)$ 在区间 $[0, 1]$ 上可导, 且 $\int_0^1 f(x) dx = 1$. 证明:

(1) 存在 $\xi \in (0, 1)$, 使得 $f(\xi) = 1$;

(2) 存在 $\eta \in (0, 1)$, 且 $\eta \neq \xi$, 使得 $\eta f'(\eta) + f(\eta) = 1$.

$$(1) \quad F(x) = \int_0^x f(t) dt$$

$$F(0) = 0 \quad F(1) = 1 \quad f(\eta) = \frac{F(\eta) - F(0)}{\eta - 0}$$

$$(2) \quad G(x) = x f(x)$$

$$G'(x) = f(x) + x f'(x)$$

$$G(0) = 0 \quad G(1) = 1$$

$$\Rightarrow G(\eta) = 1$$

1000.11.1.7

7. 证明: $\frac{2}{e} \leq \int_0^2 e^{x^2-x} dx \leq 2e^2$.

$$f(x) = x^2 - x \quad f'(x) = 2x - 1 \stackrel{\triangle}{=} 0 \Rightarrow x = \frac{1}{2}$$

x	0	$(0, \frac{1}{2})$	$\frac{1}{2}$	$(\frac{1}{2}, 2)$	2
$f'(x)$		-	0	+	
$f(x)$	0	$\searrow$	$-\frac{1}{4}$	$\nearrow$	2

$$\frac{2}{e} = \int_0^2 e^{-\frac{1}{4}} dx \leq \int_0^2 e^{x^2-x} dx \leq \int_0^2 e^2 dx = 2e^2$$

1000.11.1.8

8. 已知函数 $f(x)$ 在区间 $[a, b]$ 上连续且单调增加, 证明:

$$\int_a^b \left(\frac{b-x}{b-a} \right)^n f(x) dx \leq \frac{1}{n+1} \int_a^b f(x) dx \quad (n \in \mathbb{N}).$$

$$F(x) = \int_a^b \left(\frac{b-x}{b-a} \right)^n f(x) dx = \frac{1}{n+1} \int_a^b f(x) dx$$

$$= \int_a^b \left(1 - \frac{x-a}{b-a} \right)^n f(x) dx = \frac{1}{n+1} \int_a^b f(x) dx$$

$$F(x) = \int_a^b \left[\left(1 - \frac{x-a}{b-a} \right)^n - \frac{1}{n+1} \right] f(x) dx$$

$$G(t, F(x)) = \int_a^t \left[\left(1 - \frac{x-a}{t-a} \right)^n - \frac{1}{n+1} \right] f(x) dx$$

$$G'_t = \left[\left(1 - \frac{x-a}{t-a} \right)^n - \frac{1}{n+1} \right] f(x)$$

$$\text{证} \quad (n+1) \int_a^b (b-x)^n f(x) dx - \int_a^b (b-a)^n f(x) dx \leq 0$$

$$F(t) = (n+1) \int_t^b (b-x)^n f(x) dx - \int_t^b (b-t)^n f(x) dx$$

$$= (n+1) \int_t^b (b-x)^n f(x) dx - (b-t)^n \int_t^b f(x) dx$$

$$F'(t) = -(n+1)(b-t)^n f(t) + n(b-t)^{n-1} \int_t^b f(x) dx + (b-t)^n f(t)$$

$$= n(b-t)^{n-1} \int_t^b f(x) dx - n(b-t)^n f(t)$$

$$= n(b-t)^{n-1} \left[\int_t^b f(x) dx - f(t) \right] \geq 0$$

$$F(t) \nearrow \quad F(a) = (n+1) \int_a^b (b-x)^n f(x) dx - (b-a)^n \int_a^b f(x) dx$$

$$\leq F(b) = 0$$

1000.11.1.9

9. 设 $f(x)$ 在 $[a, b]$ 上连续, 且 $f(x) > 0$, 证明: $\ln \left[\frac{1}{b-a} \int_a^b f(x) dx \right] \geq \frac{1}{b-a} \int_a^b \ln f(x) dx$.

$$\text{证} \quad \ln \left[\frac{1}{b-a} \int_a^b f(x) dx \right] - \frac{1}{b-a} \int_a^b \ln f(x) dx$$

$$\text{设} \quad F(x) = \ln \left[\frac{1}{x-a} \int_a^x f(t) dt \right] - \frac{1}{x-a} \int_a^x \ln f(t) dt$$

$$F'(x) = \left(\frac{1}{\frac{1}{x-a} \int_a^x f(t) dt} - \frac{1}{f(x)} \right) \left(-\frac{1}{(x-a)^2} \int_a^x f(t) dt + \frac{f(x)}{x-a} \right)$$

$$= \frac{f(x) - \bar{f}(x)}{f(x) \cdot \bar{f}(x)} \left[\frac{f(x) - \bar{f}(x)}{x-a} \right] \geq 0$$

$$F(a) = 0 \quad \times$$

1000, 11.1.10

10. 设 $f(x)$ 在闭区间 $[0, 1]$ 上有二阶导数, 且 $f\left(\frac{1}{2}\right) = 1, f''(x) > 0$, 证明 $\int_0^1 f(x) dx \geq 1$.

$$\text{证} \quad \int_0^1 f(x) - f\left(\frac{1}{2}\right) dx \geq 0$$

$$\text{证} \quad \int_0^1 f(x) - f\left(\frac{1+t}{2}\right) dx \geq 0$$

$$\text{设} \quad F(t) = \int_0^t f(x) - f\left(\frac{t}{2}\right) dx$$

$$F'(t) = f(t) - f\left(\frac{t}{2}\right) > 0$$

$$F(0) = 0 \quad \ast$$